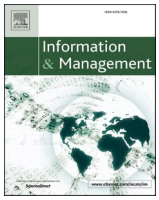
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The dark side of virtual status systems: Exploring the impact of status loss on user engagement

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ABSTRACT

In the contemporary landscape of online platforms, virtual status systems, represented by badges, titles, and ranks, play a crucial role in user identification and engagement. Prior research highlights the positive impacts of virtual status awards on user engagement. However, the potential negative effects of status loss remain underexplored. This study addresses this gap by examining the detrimental impact of virtual status loss on user engagement, emphasizing the varying intensity of the impact across multiple successive time points during users' subsequent engagements. Utilizing reinforcement theory and accountability theory, it investigates how status loss affects user engagement. The findings suggest that status loss reduces the frequency of user engagement significantly and results in a stronger short-term shock that gradually diminishes. This effect is more intense for users with newly gained status, while those who have gained status multiple times experience weaker shocks. Additionally, losing status leads to a decline in the effort and quality of users' contributions, with stronger negative impacts on engagement dimensions that users previously valued more. This study provides a nuanced understanding of the dual nature of virtual status systems, offering insights to optimize status-related user engagement strategies and enhance the development of online platforms.

1. Introduction

In the current ecosystem of online platforms, the conferral of virtual status on users has emerged as a fundamental mechanism for user identification. Virtual status systems, typically represented through badges, titles, or ranks, are strategically designed to incentivize user participation, foster community interaction, and enhance user retention [7,33]. These systems leverage foundational psychological principles, such as the desire for recognition and social standing, to motivate users to contribute content, engage in discussions, and maintain an active presence on the platform [28,44]. Prominent platforms such as Yelp, TripAdvisor, and Stack Overflow employ various forms of status symbols to highlight users' contributions, enhancing the perceived value and credibility of their content and driving business through user engagement [38]. Businesses can target users with specific statuses for campaigns, while users navigate the process of acquiring, upgrading, and potentially losing their status. Prior research suggests that status awards can significantly enhance user activity, content quality, and community engagement through positive reinforcement [9,44]. The recognition and rewards associated with status encourage users to sustain or even increase their engagement to retain prestige and social benefits [8]. In short, virtual status systems play a crucial role in the vibrancy and success of online platforms.

While prior scholars and practitioners have predominantly emphasized the positive benefits of virtual status systems for platforms and users, the potential adverse risks associated with these systems have been significantly underexplored. On platforms such as Yelp, where status (e.g., Yelp's Elite badge) is not permanent, users can experience a loss of status when they fail to meet certain standards. Despite extensive documentation of the positive impacts of virtual status systems, the existing literature on status gain provides limited insights into the negative repercussions that may arise from status loss. Intuitively, status loss might appear as a natural counterpart to status gain; however, studies in offline contexts (e.g., airline frequent-flyer programs and travel points) demonstrate that losing or downgrading status can lead to customer attrition [3,37]. In an online platform context, users experiencing status loss may suffer from diminished motivation, leading to reduced engagement or even complete disengagement. These risks can result in decreased user retention and increased user attrition. These counterproductive effects are critical, as the demotivating consequences

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0378-7206/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

of status loss can undermine overall user engagement and the health of online communities, compromising a platform's ecosystem. There is a substantial gap in the literature regarding the behavioral consequences of status loss in online platforms, warranting urgent scholarly attention.

This study aims to address this research gap by meticulously examining the impact of virtual status loss as a negative reinforcer on user engagement. Grounded in reinforcement theory and accountability theory, it formulates hypotheses and conducts analyses to explore several key aspects. It investigates the negative effect of status loss on the frequency of user engagement and provides an in-depth analysis of how this effect evolves, highlighting its initial intensification and subsequent gradual diminishment over time. Additionally, it examines how users' past status experiences influence the extent to which their engagement is affected by status loss. It also analyzes how the loss of status impacts users' performance, specifically their effort and the quality of their contributions. Furthermore, it explores the directional changes in users' engagement behavior, revealing that the dimensions of engagement most valued by users prior to status loss are more negatively affected.

By focusing on the changes in user behavior following status loss, this study provides a comprehensive understanding of the double-edged nature of virtual status systems. This exploration contributes to a more balanced perspective on the role of virtual status in user engagement, informing the development of differentiated status awarding, promotion, and deprivation strategies tailored to individual user characteristics (e.g., historical engagement patterns and past status experiences). Optimizing user status management mechanisms within virtual status systems can enhance overall user engagement, promoting the prosperity and health of the platform community.

The structure of this paper is as follows. We begin with a comprehensive review of the theoretical background and literature related to the influence of virtual status. This is followed by the development of our research hypotheses. Subsequently, we describe the data context and the operationalization of our variables. We then present the empirical analysis and results. Finally, we discuss the theoretical and managerial contributions of our findings and outline potential avenues for future research.

2. Conceptual and literature background

2.1. Reinforcement theory

In the realm of management practice, reinforcement theory (referred to by some as operant conditioning) emphasizes that an individual's behavioral motivation is intricately linked to the consequences of their past actions [32]. Variations in an individual's psychology or behavior often stem from disparities in the outcomes of previous actions or shifts in environmental factors [34]. Generally speaking, positive outcomes serve as positive reinforcers, bolstering an individual's subsequent behavioral motivation and increasing the frequency of relevant actions. Conversely, negative outcomes act as deterrents (negative reinforcers), diminishing an individual's subsequent motivation and reducing the frequency of their behavior [5].

In the specific context of our study, the virtual badge serves as a symbol of status, influencing user engagement on the platform [2,44]. Gaining status as a positive consequence of prior engagement behavior strengthens users' motivation to further engage, while the loss of status weakens their motivation [33]. In this study, we utilize the gain or loss of status as positive or negative reinforcers, respectively, to analyze how these stimuli shape user engagement behavior on the platform.

2.2. Accountability theory

Accountability theory asserts that the perceived obligation to justify one's actions to others instigates a sense of accountability for individuals regarding their behavior and decisions [35]. As a behavioral mechanism, this theory elucidates how individuals adapt their behavioral motivations in response to both intrinsic and extrinsic stimuli [44]. Previous research emphasizes that factors such as the presence of others and the anticipation of evaluations bolster individuals' accountability [18]. In studies by Vance et al. [36], it was observed that systematic manipulation of identifiability, evaluation expectation, monitoring awareness, and social presence forms the foundation of accountability theory in explaining individual behavior within information systems. The relative significance attributed to these factors by users influences their behavioral inclinations within such systems.

Within the context of online review platforms, users who gain status perceive a heightened level of identifiability and social presence [44]. Also, they become more aware of being evaluated and monitored by others, leading to a strengthened sense of accountability for their actions. Conversely, the loss of status diminishes users' accountability for their behavior [33]. Thus, the current study will propose hypotheses by analyzing the fluctuations in users' perceptions of the manipulated elements of accountability theory before and after experiencing a loss of status.

2.3. Effects of virtual status on users' platform engagement motivation

In contemporary online platforms, users' statuses are commonly classified into hierarchical titles and non-hierarchical badges. Past research has explored the effects of these awards on user motivation to engage with platforms (a summary is given in Table 1). Hierarchical status has been found to influence users' motivation to engage positively, yet motivation tends to decline after an upgrade is achieved [12]. Gaining non-hierarchical badges can also motivate users to engage, but this effect is observed primarily among users with moderate or low motivation and is effective only in the short term [8,44]. Furthermore, higher initial motivation, user reputation, and repeated status acquisition can weaken the positive effect of status on engagement [2,5,9].

While most studies have focused on the positive effects of status gains on user engagement, research on the impact of status loss is extremely limited. Tamaddoni et al. [33] conducted a preliminary study on status loss within a review platform, finding a negative impact on the quantity, valence, and elaborateness of reviews, with less experienced users being more negatively affected. In an earlier study, Mutter and Kundisch [25] examined system-wide status loss due to platform restructuring in a Q&A community, finding that substantial status loss can significantly reduce user activity levels.

Overall, existing literature lacks a detailed theoretical analysis of status loss's impact, particularly regarding its long- and short-term effects on user engagement and multifaceted contribution characteristics across different levels of user motivation and status acquisition experiences. Addressing this gap, the current study makes the following contributions. First, it advances understanding of status loss within online platform engagement, offering a more balanced perspective on how virtual status influences user contributions. Second, it reveals how status loss affects specific engagement dimensions—frequency, effort, and quality—highlighting that motivation decline is uneven and more pronounced in areas users prioritize most. Finally, it identifies the moderating effect of status acquisition experience, showing that losing newly gained status (vs. repeatedly gained) leads to a greater reduction in engagement.

2.4. Status loss in offline context

Although research on the impact of status loss within online platforms is limited, its effects have been explored in some offline contexts. Studies of loyalty programs, such as airline frequent-flyer programs, have explored how customer status loss affects loyalty intentions toward a firm. Tamaddoni et al. [33] found that status loss has a weaker negative impact on engagement than the positive effect of status gains. In contrast, Wagner et al. [37] observed that status loss exerts a negative

Table 1
Studies on the effect of virtual status on users' platform engagement motivation.

Study	Context	Theory applied	Status		Related findings
			Changes	Forms	
Mutter and Kundisch [25]	Q&A community	Status-seeking behavior	Loss	Hierarchy	The system-wide loss in status elicits negative behavioral responses from users, particularly among those experiencing substantial status declines.
Goes et al. [12]	A knowledge exchange website	Goal setting theory, status hierarchy theory	Gain	Hierarchy	Status incentives initially boost user contributions, but contribution levels drop significantly once status is gained. This positive effect on contribution appears temporary and decreases for higher status levels.
Chen et al. [9]	StackExchange	Dynamic motivation/ contribution in online communities	Gain	Badge	Gaining status will motivate users with low and medium engagement motivation to transition to the high motivation state, but it will not offer further incentives to those already in the high motivation state.
Zhang et al. [44]	Yelp	Accountability theory	Gain	Badge	In the short term, gaining status increases reviewers' contributions, promotes more conservative grading, and enhances review readability. In the long term, reviewers with status maintain review quality and stable rating behavior without increasing their contributions.
Bhattacharyya et al. [5]	Yelp	Reinforcement theory	Gain	Badge	Gaining status initially heightens contribution behavior, but multiple status gains may result in reduced contributions due to satiation. Additionally, deserving members overlooked for status may contribute less due to feelings of unfairness.
Chen et al. [8]	StackExchange	Goal setting theory, expectancy-value theory	Gain	Hierarchy & badge	The relationship between proximity to higher hierarchical status and sustained knowledge contribution is nonlinear. Non-hierarchical status also fosters knowledge contribution. User reputation diminishes the influence of both hierarchical and non-hierarchical status on future contributions.
Banerjee et al. [2]	Yelp	Intrinsic/extrinsic motivation theory	Gain	Badge	The effect of gaining status on contribution is influenced by the aspect of contribution and the individual's experience with status acquisition.
Tamaddoni et al. [33]	Yelp	Accountability theory, functional role theory, and prospect theory	Gain & Loss	Badge	The positive effect of gaining status on the number of reviews outweighs the negative impact of losing status, particularly among less experienced reviewers.
Current study	Yelp	Reinforcement theory, accountability theory	Loss	Badge	Status loss reduces user engagement frequency, with a stronger short-term impact that gradually diminishes. Newly gained status users experience more intense effects, while repeated status users face weaker shocks. Status loss leads to a decline in effort and quality, with stronger negative impacts on previously valued engagement dimensions.

impact on loyalty intentions stronger than the positive impact associated with status gains. Banik et al. [3] further emphasized that the negative impact of status loss, driven by feelings of frustration and social discomfort, is particularly pronounced among long-term and female members.

Applying insights from loyalty programs to online platforms with virtual rewards presents unique challenges, as offline status is typically linked to financial or consumption-based actions. In contrast, status on online platforms is primarily achieved through non-monetary activities like content contributions and peer interactions. Moreover, status on review platforms is publicly visible, unlike the private status common in most loyalty programs. Therefore, our study diverges from traditional, consumption-based status systems to focus on the negative reinforcement effects of status loss within a public, non-monetary setting.

3. Hypothesis development

The most salient facet of user engagement with a platform resides in the variation of motivational impetus, quantifiable through frequency of engagement [2,5,16,33,40]. This study begins by examining the dynamics of users' engagement frequency following a loss of status, along with its temporal patterns. Additionally, we explore the moderating effect of users' previous experiences related to status. As engagement dynamics also evolve when users remain active, we subsequently investigate shifts in the effort and quality dimensions of users' engagement. Ultimately, we analyze the directional changes in various dimensions of users' engagement performance.

3.1. Engagement frequency

Drawing from reinforcement theory, status functions as a social reinforcer for individuals [5], motivating users' engagement when they earn the status. Apart from the observed increase in user engagement resulting from elevated status, the loss of status can have a counterproductive effect, undermining users' motivation to engage. When users experience a reduction in status or social recognition, they may perceive reputational harm, thus diminishing their motivation to contribute to the online community [4]. This shift in status triggers a reverse effect akin to negative reinforcement, resulting in users' decreased motivation to engage with the platform [5].

Accountability theory provides a more granular theoretical mechanism for the negative reinforcement effect of status loss. Users who gain status are likely cognizant of being monitored and of the social presence within the platform [39] Wang et al., 2021; [44]. As a result, they tend to demonstrate greater accountability, which leads to heightened engagement. Concurrently, the loss of status diminishes the user's perception of being monitored and their sense of social presence, leading to a decrease in their sense of accountability on the platform [33]. Consequently, the users' motivation to engage with the platform becomes more negative following the loss of status compared to before. In other words, users who have lost their status are less inclined to engage with the platform than they were prior to the loss. Therefore, we hypothesize that

H1: Users who lose status will exhibit a reduction in their platform engagement frequency.

Moreover, the positive or negative social reinforcement of individual behavioral motivation stemming from gaining or losing status is considered short-lived [12]. Research has indicated that the

reinforcement of engagement behavior by users who have gained a badge symbolizing status diminishes over time [44]. Conversely, it can be argued that the negative impact of losing status on users' platform activity performance also diminishes with the passage of time. In other words, we contend that users experience a rapid decline in motivation for platform engagement after losing status. As time progresses and users' psychological distance from the shock of status loss increases, the negative reinforcement of engagement activities by this event weakens.

Accountability theory also provides a consistent prediction regarding status changes. Research suggests that status changes initially deliver a strong shock to users' accountability and motivation; however, a possible "complacency" effect may cause the intensity of these responses to diminish over time [44]. Specifically, when users encounter a status loss, they initially undergo a strong shock in their sense of social presence and awareness of monitoring, resulting in an immediate reduction in accountability for actions such as writing reviews. Nevertheless, a single instance of status loss does not sustain a prolonged accountability shock; users gradually adapt, resulting in a reduced response in their decreased engagement frequency over time. Therefore, we propose the following hypothesis:

H2: The reduction in engagement frequency is most pronounced in the initial period following the loss of status, and diminishes over time.

The loss of status can significantly dampen a user's motivation to engage, and the extent of this impact can correlate with their pre-existing level of engagement motivation before encountering this negative reinforcement (i.e., prior to the status loss). Bhattacharyya et al. [5] illustrated that newly acquired virtual status yields the most substantial social reinforcement for individuals, while repeated status rewards may lead to diminishing returns in reinforcement. Those who newly gained status were likely most susceptible to the reinforcing effect of status on their engagement motivation, surpassing the influence observed in users who had gained status repeatedly. This can be explained by the satiation effect, where individuals become accustomed to a repeated stimulus, leading to diminished responsiveness [11]. Repeated status experiences reduce their sensitivity to evaluative expectations, enhancing their adaptability to status changes. As a result, users with repeated status gains become less sensitive to status loss as repeated awards lose their perceived value.

An alternative perspective arises from loyalty program literature: that long-term customers are more likely to perceive status demotions as more intensely unfavorable than short-term customers and will also experience stronger feelings of frustration and social discomfort [3]. However, loyalty program status is closely tied to monetary spending and privileges, with status loss directly impacting a customer's financial benefits. This makes the intensity of a customer's reaction highly dependent on their financial investment, as losing status typically equates to losing significant benefits.

In contrast, status in this study is represented by platform-awarded badges, which hold primarily symbolic value with minimal monetary significance. Unlike loyalty program status, losing these badges does not result in tangible financial consequences. Additionally, alongside the explicit badge-based status, users also hold implicit status, such as a fan base, which tends to increase with repeated status gains. High implicit status weakens the negative impact of explicit status loss, contrary to findings in loyalty program research. This mitigation occurs because users with higher implicit status retain a heightened sense of identifiability, monitoring, and social presence, making them less affected by status loss than users with newly gained status. For badge-based status, Tamaddoni et al. [33] found that more experienced reviewers are less influenced by external status symbols and instead rely more on their own experience. This aligns with our argument that the accumulation of implicit status mitigates, rather than amplifies, the negative reinforcement effect of status loss on engagement frequency. Therefore, we hypothesize that

H3: Users who lose newly gained (vs. repeatedly gained) status will exhibit a greater reduction in their platform engagement frequency.

Aside from whether a user has gained status repeatedly, the cumulative number of times a user has gained status on a platform will also mitigate the negative effect of status loss on a user's engagement. In comparison with users who have accumulated numerous status gains, those with fewer status gains will receive a heightened level of reinforcement for their engagement motivation. Additionally, users with fewer status gains may exhibit heightened sensitivity to the loss of explicit status conferred by badges, as they are likely to lack sufficient accumulated implicit status (e.g., followers) to counterbalance the perceived significance of this explicit status. Consequently, users with fewer times of status gain will exhibit a more pronounced decline in motivation to engage upon losing their status than those with more frequent status gains. And we hypothesize that

H4: When losing status, users who had gained fewer (vs. more) times of status will exhibit a greater reduction in their platform engagement frequency.

3.2. Effort and quality

We have delved into how users' motivation to engage with a platform declines following the loss of their status. However, this decline is not confined to the presence or absence of engagement activities (i.e., frequency), but also extends to the effort and quality of those activities. Even for ongoing activities, users' engagement performance can fluctuate.

In terms of motivational reinforcement, users' willingness to participate diminishes upon losing their status compared with when they possess it. Consequently, if a user decides to continue contributing, their reduced engagement willingness manifests in decreased effort and quality in their engaging behavior. Aligned with accountability theory, the user's awareness of monitoring and social presence diminishes after status loss, reducing their level of accountability for their engagement performance to some extent. This reduced sense of accountability will be reflected in the substance of users' contributions, as they perceive a reduced need to adhere to previous behavioral standards. Consequently, as a user interacts with a platform amid this decline in accountability, the effort invested in and the quality of the content contributed during engagement are likely to suffer compared to before the status loss. Therefore, we hypothesize that

H5a: Users who lose status will exhibit a reduction in their platform engagement effort.

H5b: Users who lose status will exhibit a reduction in their platform engagement quality.

3.3. Directionality of changes in different behavior dimensions

Accountability theory suggests that heightened accountability leads to a greater inclination to explain and regulate one's behavior, driven by increased awareness of monitoring and social presence. Furthermore, people tend to prioritize various dimensions of their behavior, such as frequency of engagement, level of effort, and quality of contribution, based on perceived social expectations [29]. In our context, users who gain status on a platform tend to allocate their attention to three primary dimensions of engagement behavior, frequency, effort, and quality, guided by their interpretation of social norms within the platform's specific community. Different users may prioritize these dimensions differently upon gaining status. For example, some users, following their perceived platform social norms, might prioritize maintaining a high frequency of engagement to sustain their status. Conversely, others might emphasize investing significant effort in their contributions or ensuring high-quality outputs. Consequently, excelling in one dimension, such as engagement frequency, does not necessarily imply excellence in others, like effort or quality.

An individual's preference reflects their self-efficacy for the preferred engagement dimension, fostering beliefs in their capability to achieve particular goals through actions within that domain [1,21].

However, experiencing a loss of status can diminish motivation. This preference may subsequently lead the individual to perceive low self-efficacy in that domain [1], resulting in a decline in performance within their prioritized dimension of engagement [30]. For instance, a user who previously emphasized effort in their interactions may reduce their efforts after losing status. In contrast, another user, who initially did not prioritize effort, may exhibit minimal adjustment in their level of behavioral effort following a status loss, reflecting a perception of effort as less consequential. Notably, in this scenario, the user who previously demonstrated greater engagement effort compared with those who perceive effort as less consequential tends to exhibit a more pronounced drop in effort following a status loss. This highlights a directional shift in engagement behavior across different dimensions before and after status loss. That is, users' varying prioritization of engagement dimensions such as frequency, effort, and quality enables them to make differential adjustments to these dimensions after losing status, with those dimensions that were previously prioritized being more significantly adjusted. Therefore, we formulate the following hypotheses:

H6a: Users who previously had higher (vs. lower) platform engagement frequency will exhibit a higher reduction in their platform engagement frequency after losing their status.

H6b: Users who previously had higher (vs. lower) platform engagement effort will exhibit a higher reduction in their platform engagement effort after losing their status.

H6c: Users who previously had higher (vs. lower) platform engagement quality will exhibit a higher reduction in their platform engagement quality after losing their status.

Fig. 1 illustrates the primary factors under investigation in our research model.

4. Research methodology

4.1. Data and context

In this study, we utilize Yelp as our research platform. Yelp is one of the largest online review systems. At the beginning of each year, Yelp selects a group of users to join the Yelp Elite Squad, awarding them the Elite badge. Users can be nominated by themselves, other users, or local community managers. They are evaluated based on their engagement with the platform over the past year, considering factors such as quality reviews, high-quality tips, detailed profiles, positive voting and liking history, and overall interactions. The badge remains valid until the end of the calendar year, after which all users are renominated and reassessed for the following year.

When a user receives the badge, it is displayed on their profile next to their avatar, enhancing their reputation. In this study, the Elite badge serves to differentiate user status: Those awarded the badge in the current year have gained status, while those awarded the badge in the previous year but not in the current year have lost status. Users who lose status may have held it for multiple years consecutively or intermittently, or it may be newly acquired status that was not awarded in the year before the previous one.

For our empirical analysis, we use open datasets from Yelp tailored for academic research and educational purposes (https://www.yelp.com/dataset), which are extensively utilized by scholars [5,23,26,27,33,41]. These datasets offer rich information about reviewers, reviews, users, and businesses, supporting empirical research on user behavior dynamics on e-commerce platforms. Our original dataset comprises 6685,900 online reviews contributed by 1637,138 users for 192,606 businesses in the United States and Canada, spanning from October 2004 through November 2018.

Yelp provided all reviews for the focal businesses in these open data prior to the data release deadline, though some users' reviews (for businesses not included in the sample) were not all included. For analysis purposes, we adopted Bhattacharyya et al.'s [5] approach and selected a study sample of 625,317 users whose reviews in the Yelp open dataset accounted for more than 70 % of their total actual reviews, as also reported by Yelp's open dataset. We focused on the review engagement behavior of these users in 2016 and 2017, the years with the largest numbers of reviews with complete records (January through December). We excluded users who registered during the sample period and those who did not post a review in 2016. Users who did not post a review in 2017 were retained, as those who lost their status might have been less motivated to engage. Additionally, we will present results excluding users who did not post reviews in either 2016 or 2017 in our robustness analysis.

Ultimately, our sample includes 200,172 users. Of these, 491 users lost status in 2017 (Elite in 2016 but not in 2017), 974 users gained status newly in 2017 (not Elite in 2016 but Elite in 2017), 1946 users gained status repeatedly in 2017 (Elite in both 2016 and 2017), and 196,761 users had no status (Elite neither in 2016 nor in 2017).

4.2. Propensity score matching

There may be self-selection or time-shock-induced endogeneity between users' behavioral performance after losing status and the event of losing status. For example, comparing users who have lost status with those who have newly gained it might reveal a decrease in motivation for the former due to increased motivation for the latter. Additionally, comparing users who have lost status with those who remain statusless might produce results influenced by the fact that statusless users, often lurkers, are less affected by time-related external shocks. Therefore, any observed differences could be due to stronger external shocks impacting the once-active users who lost their status rather than the lurkers. To address this endogeneity issue, we use the propensity score matching (PSM) method to identify control group users who match the treatment group (users who lost status) in our study.

Specifically, Yelp does not explicitly assess whether a user can enter the Elite Squad; users who lose status may regain it by improving their performance slightly, while those who repeatedly gain status may lose it by underperforming slightly. Thus, we can find a control group with similar engagement behavior patterns among repeat status gainers for those who lost status. We employ a Logit model to regress a binary indicator of whether a user has lost status (considering users who have lost status and those who have repeatedly gained status) based on their platform engagement characteristics in 2016 (Table 2). We then compute the users' propensity scores and use the *k*-nearest-neighbors matching method to perform 1:1 matching of the treatment group with a control group.

We present the sample differences between the treatment and control groups after matching in Table 3. The standardized deviations (% bias) of all variables after matching are less than 10 %, indicating that our matching process successfully creates balanced samples between the treatment and control groups. Additionally, Fig. 2 shows the propensity scores of the treatment and control groups before and after matching, further illustrating the comparability of the groups post-matching.

4.3. Measures

Posting online reviews is significant engagement behavior among users on online platforms. In this paper, we explore how users' platform engagement behavior—specifically, the frequency, effort, and quality of posting reviews—changes following a loss of status. Since the users' status changes annually, our study integrates data primarily at the yearly level. Additionally, to examine the time-varying nature of users' platform engagement characteristics, we process data at the monthly level.

Dependent variables. We measure the frequency of a user's platform engagement using the yearly number of reviews posted by a user (*Rev_N*). The engagement effort is measured by the annual average length of a user's reviews (*RevLen*), which is the average number of characters per review [20]. Given that the number of peer votes received

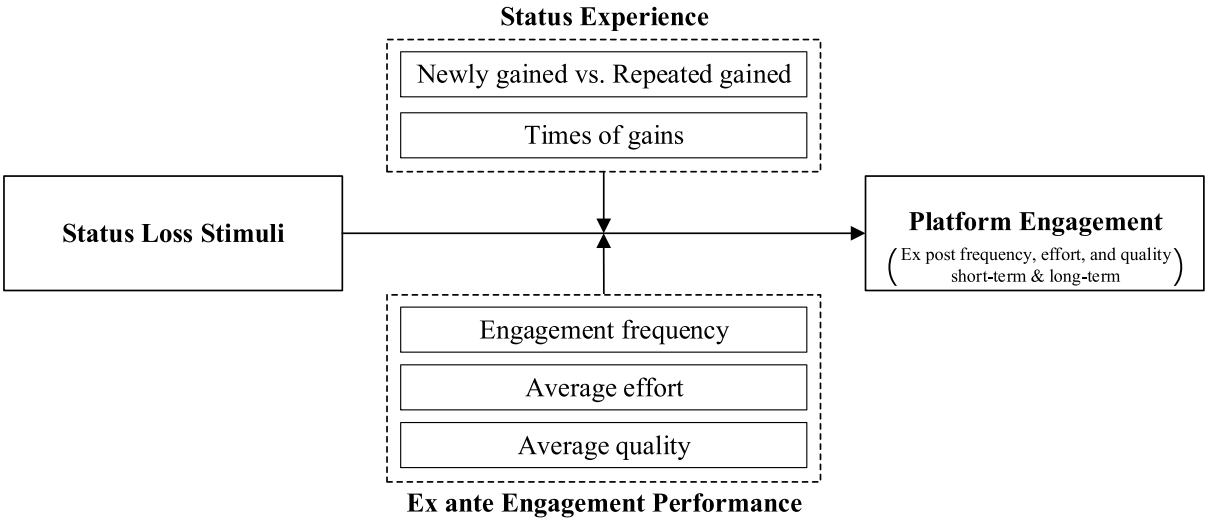


Fig. 1. Research model.

Table 2
Variables employed in the PSM process.

Variable	Description	Mean	SD	Min	Max
Rev_N	Number of reviews	29.50	29.93	0	366
RevFunVotes	Number of reviews' funny votes	37.45	143.00	0	3657
RevUseVotes	Number of reviews' useful votes	87.38	236.20	0	4958
RevCoolVotes	Number of reviews' cool votes	56.00	200.50	0	4663
RevLen_AVG	Average character number in reviews	861.80	454.30	0	3924
RevRating_AVG	Average rating of reviews	3.84	0.49	1	5
Tip_N	Number of tips	6.16	17.31	0	391
TipLikes	Number of likes received by tips	0.20	1.10	0	20
TipLen_AVG	Average number of characters in tips	43.68	50.18	0	456
Cumu_Revs (log)	Cumulative number of reviews before the focal year	3.77	1.36	0	7.27
Cumu_Tips (log)	Cumulative number of tips before the focal year	1.77	1.65	0	6.73
Cumu_badges	Cumulative number of badges before the focal year	1.36	1.68	0	8
Tenure (log)	Duration of user registration on the platform, measured in months.	3.83	0.50	2.64	4.96

Note. The focal year for the variable is 2016.

by users can serve as an indicator of the quality and informativeness of their contributions [6,14,17,19], we assess their engagement quality by using the user's yearly average number of votes received for all reviews (*RevVote*), including useful, funny, and cool votes.

Independent variables. Our theoretical framework focuses on the event of a user's loss of status on the platform. We use the dummy variable *Lose* to indicate whether a reviewer belongs to the treatment group (those who lost status) or the control group (repeat status gainers matched to the treatment group). The dummy variable *Post* represents the experimental period, with 2016 coded as 0 and 2017 (when the treatment group lost status) coded as 1.

To investigate the impact of a user's status experience, we introduce the dummy variable *NewGain*, which indicates whether the status lost had been gained newly or repeatedly. This variable is set to 1 if the user's status in 2016 was newly gained (i.e., they had no status in 2015). We also use the count variable *Badges*, which counts the number of times the user had been awarded an Elite badge prior to 2016. This allows us to explore how a user's status experience influences the impact of losing status on their platform engagement behavior.

Table 3
Differences between treatment and control groups after matching.

Variable	Mean		%bias
	Treated	Control	
Rev_N	12.27	14.58	−9.5
RevFunVotes	11.02	12.93	−1.6
RevUseVotes	29.62	35.30	−2.9
RevCoolVotes	16.75	19.44	−1.6
RevLen_AVG	860.80	865.50	−1.0
RevRating_AVG	3.80	3.86	4.8
Tip_N	2.17	2.10	0.5
TipLikes	0.05	0.06	−1.6
TipLen_AVG	30.28	31.53	−2.5
Cumu_Revs (log)	3.85	3.81	2.7
Cumu_Tips (log)	1.63	1.57	3.8
Cumu_badges	1.43	1.39	2.5
Tenure (log)	3.86	3.86	−0.2

Control variables. We control for individual users' historical engagement performance, social performance, and platform engagement seniority, as these factors may influence their review behavior. Specifically, we control for the total number and average rating of a user's past reviews [45], the size of their social connections on the platform [43], and the length of time they have been registered on Yelp. Additionally, we control for the user's city of residence, represented by dummy variables based on the user's personal location, as displayed on their Yelp pages. All variables of interest are described in Table 4.

5. Model specification and results

To examine the impact of experiencing a loss of status between two annual periods on user engagement, we employ a difference-in-differences (DID) model for empirical analysis. This model, applied after matching treatment and control group samples via PSM, effectively identifies the impact of the status loss event on user behavior.

5.1. Engagement frequency

In investigating the impact of status loss on user engagement frequency, we first introduce an interaction term between the variable *Lose*,

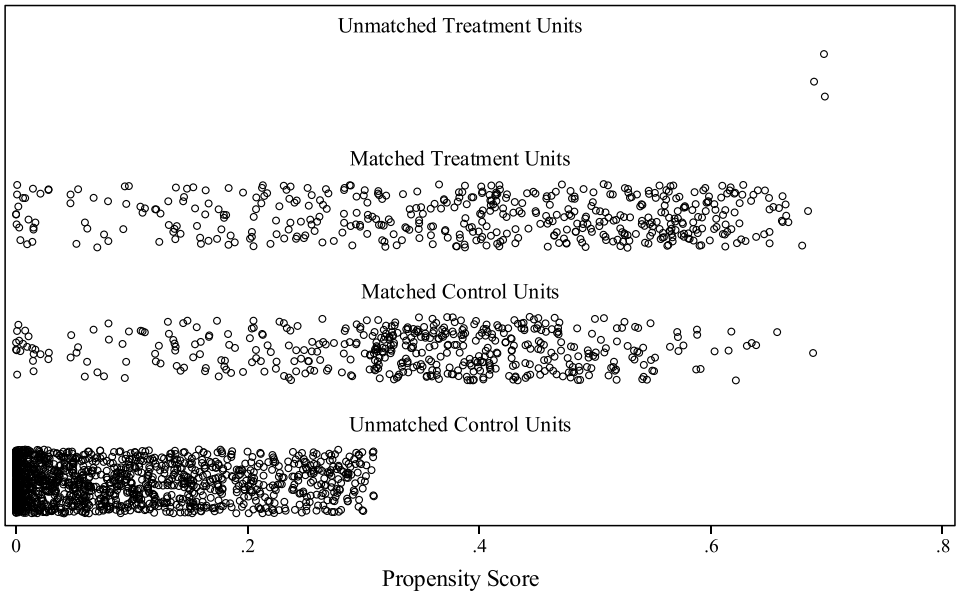


Fig. 2. Distribution of propensity scores after matching.

Table 4
Variable definitions and descriptive statistics.

Variable	Definition	Mean	SD	Min	Max
Dependent Variables					
$Log\#Rev_N_{it}$	Log number of reviews posted by user i in year t	2.02	1.04	0	4.98
$Log\#RevLen_{it}$	Log of average number of characters in reviews posted by user i in year t	5.89	2.11	0	8.31
$Log\#RevVote_{it}$	Log of average number of votes for reviews posted by user i in year t	1.20	0.81	0	5.33
Independent Variables					
$Lose_i$	Whether user i is in the treatment group: yes = 1, otherwise 0	0.50	0.50	0	1
$Post_t$	Whether year t is in the post-event period: yes = 1, otherwise 0	0.50	0.50	0	1
$NewGain_i$	Whether user i's status was newly gained in 2016	0.35	0.48	0	1
$Badges_i$	Log number of Elite badges gained by user i prior to 2016	1.91	1.66	0	9
Control Variables					
$Log\#CumRevs_{it}$	Log number of reviews posted by user i prior to year t	4.00	0.89	0	7.04
$AvgRating_{it}$	Average rating of reviews posted by user i prior to year t	3.76	0.48	0	5
$Log\#SocialSize_{it}$	Log of total number of friends and fans of user i at the beginning of year t	4.30	1.27	0	8.94
$Tenure_{it}$	Number of years user i has been registered on Yelp at the beginning of year t	3.78	1.94	0.08	10.83

Notes. Rev_N_{it}, $RevLen_AVG_{it}$, $RevVote_AVG_{it}$, $CumRevs_{it}$, and $SocialSize_{it}$ are log-transformed as their highly skewed distributions. Given that some variables have a minimum value of zero, we use $\log(V + 1)$ when transforming these variables.

representing the event shock (distinguishing between treatment and control groups), and the variable *Post*, representing the experimental phase (distinguishing before and after the event), to test H1. Building upon this, we introduce the variables *NewGain* and *Badges* as interaction terms to further examine the moderating effect of users' status experiences on the impact of status loss (testing H3 and H4). Our model is formulated as

$$Log\#Rev_N_{it} = \alpha + \beta_1 Lose_i + \beta_2 Post_t + \beta_3 Lose_i \times Post_t + \beta_4 Lose_i \times Post_t \times NewGain_i + \beta_5 Lose_i \times Post_t \times Badges_i + \gamma * Control_{it} + \varepsilon_{it}, \quad (1)$$

where i denotes user i, t denotes period t, and $Control_{it}$ includes the control variables as well as the double interaction terms within the triple interaction term, which are of limited relevance to the analysis and are therefore merged into the control variables. Our examination centers on the coefficients β_3, β_4, and β_5 to test H1, H3, and H4, respectively.

We begin by analyzing the overall impact of losing status on user engagement frequency (H1, H3, and H4), with the temporal dynamics of this effect (H2) slated for further empirical scrutiny in subsequent analyses. The findings are detailed in Table 5. The results in column 2 of the table reveal a significant negative effect of status loss on users' frequency of review contributions ($b = -1.078$, $p < 0.001$), providing support for H1. This implies that, statistically, users' frequency of reviewing after losing status dwindles to approximately 29 %[1] of what it would have been had they retained their status, indicating a substantial decline in their willingness to engage with the platform. Further, we plot the predictive margins of the effect of losing status (or not) on user engagement frequency in Panel (a) of Fig. 3.

Subsequently, we introduce interaction terms of *NewGain* and *Badges* for the model's main effect of *Post* × *Lose* to examine how users' prior status experiences modulate the impact of losing status. The coefficient associated with the triple interaction term involving *NewGain* in column 3 is significantly negative ($b = -0.278$, $p < 0.05$), suggesting that users who have newly gained status show a more substantial decline in platform engagement following status loss, supporting H3. Panel (b) of Fig. 3 illustrates the predictive margins of status loss, contingent on whether the status was newly gained. The predictive margins indicate that users with newly gained status show a sharper engagement decline,

[1] The calculation is as follows: $\frac{Rev_N^{lose}+1}{Rev_N^{repeat}+1} = e^{-1.078}$. Setting Rev_N^{repeat} to 13.4, the average review number in 2016, gives $Rev_N^{lose} = 3.9$, resulting in 3.9/13.4 = 29%. The "+1" is from our logarithmic transformation.

Table 5
Effects of status loss on user engagement frequency.

	DV: *Log#Rev_N*			
	(1)	(2)	(3)	(4)
Post		−0.097*	−0.021	−0.129
		(0.044)	(0.055)	(0.073)
Lose		−0.293***	−0.393***	−0.292***
		(0.048)	(0.057)	(0.063)
Post × *Lose*		−1.078***	−1.014***	−1.330***
		(0.064)	(0.077)	(0.115)
Post × *Lose* × *NewGain*			−0.278*	
			(0.139)	
Post × *Lose* × *Badges*				0.103*
				(0.041)
Post × *NewGain*			−0.197*	
			(0.085)	
Lose × *NewGain*			0.333***	
			(0.094)	
Post × *Badges*				0.029
				(0.027)
Lose × *Badges*				0.000
				(0.028)
NewGain	0.236***	0.266***	0.287***	0.269***
	(0.071)	(0.061)	(0.072)	(0.061)
Badges	−0.162***	−0.082**	−0.089***	−0.127***
	(0.027)	(0.025)	(0.025)	(0.029)
CumRevs	0.065	0.052	0.083	0.076
	(0.049)	(0.043)	(0.044)	(0.043)
AvgRating	−0.279***	−0.208***	−0.192***	−0.208***
	(0.048)	(0.043)	(0.041)	(0.043)
SocialReach	0.129***	0.059**	0.054**	0.056**
	(0.022)	(0.021)	(0.021)	(0.021)
Tenure	−0.012	−0.003	−0.005	−0.004
	(0.017)	(0.014)	(0.014)	(0.014)
City FE	YES	YES	YES	YES
N	1892	1892	1892	1892
R^2	0.133	0.421	0.429	0.427

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

potentially due to stronger motivational reinforcement from recent status gains, as posited in H3.

Conversely, the coefficient linked to the triple interaction term involving *Badges* in column 4 is significantly positive ($b = 0.103$, $p < 0.05$), suggesting that users with a history of frequent status gains exhibit a diminished decline in platform engagement subsequent to losing status. In essence, individuals with fewer status gains are more likely to experience a pronounced dip in engagement after losing status, supporting H4. Panel (c) of Fig. 3 illustrates the predictive margins of status loss, contingent on high versus low levels of *Badges*. The predictive margins indicate that users with lower status experience have a sharper engagement decline, potentially due to stronger motivational reinforcement from infrequent status gains, as posited in H4.

Next, we examine the temporal shift in the frequency of user engagement following the loss of status, as posited in H2. We extend the temporal scope of our analysis from two annual periods to 24 months. We quantify the monthly frequency of user reviews, denoted as Rev_N^{Month}, representing the number of reviews posted by users each month. The month variable (January through December) is transformed into 12 categorical variables (dummy variables), labeled as *Month*_1 to *Month*_12. To validate H2, which suggests a stronger short-term effect immediately following the loss of status, we designate *Month*_12 as the base period. Subsequently, we interact *Month*_1 to *Month*_11 with the *Post* × *Lose* interaction term to examine variations in the effect of *Post* × *Lose* on the frequency of users' platform engagement across each month. The model is specified as follows:

$$Log\#Rev_N_{i\tau}^{Month} = \alpha + \beta_1 Lose_i + \beta_2 Post_\tau + \beta_3 Lose_i \times Post_\tau + \beta_4 Lose_i \times Post_\tau \times Month_\boldsymbol{X}_\tau + \gamma * Control_{i\tau} + \varepsilon_{i\tau} \quad (2)$$

where τ denotes month τ, *Month_**X*** represents the 11-month categorical variables ranging from *Month*_1 to *Month*_11, and the subvariables and sub-interaction terms of the triple interaction are incorporated into the control variables.

Column 1 of Table 6 presents the results. Particularly, the greatest negative moderating effect is observed in month 2, which is significantly greater than in months 4 through 12. We illustrate these coefficients in Fig. 4, where it is evident that the impact of losing status on the frequency of user review contributions is most pronounced in the first three months and gradually diminishes in subsequent months. This finding supports H2, indicating that the negative effect of losing status on user platform activity is most acute immediately after the event and attenuates over time. To distinguish more clearly between the short- and long-term effects of status loss, we group the monthly dummies into quarterly and semiannual periods. The re-estimated results, shown in Appendix A, consistently and more explicitly demonstrate the stronger short-term impact of status loss.

Additionally, we introduce the variable *NewGain* as an interaction term, resulting in a quadruple interaction term detailed in column 2 of Table 6. It can be observed that if the status lost by the user was newly gained in the last year, the *NewGain* feature amplified the negative impact of status loss on user commenting behavior in months 2 and 3. The lack of significance in month 1 may be attributed to users' uncertainty about whether they have lost their status (as this event occurs in January), with some still hopeful of regaining status. However, once users confirm their loss of status in month 2, they quickly exhibit negative changes in their engagement with the platform.

5.2. Effort and quality

To assess the impact of the loss of status event on the level of contribution effort and quality during user platform engagement, we introduce *Log#RevLen* and *Log#RevVote* as dependent variables in the model:

$$Log\#RevLen_{it} = \alpha + \beta_1 Lose_i + \beta_2 Post_t + \beta_3 Lose_i \times Post_t + \gamma * Control_{it} + \varepsilon_{it} \quad (3)$$

$$Log\#RevVote_{it} = \alpha + \beta_1 Lose_i + \beta_2 Post_t + \beta_3 Lose_i \times Post_t + \gamma * Control_{it} + \varepsilon_{it} \quad (4)$$

Table 7 displays the regression results. In both models, the coefficient of *Post* × *Lose* is statistically significant and negative, suggesting that the loss of status leads to reduced effort in platform engagement and lower quality content production, thereby supporting H5a and H5b. Furthermore, given the log transformation of the dependent variable, the results indicate that users' engagement effort decreases more noticeably than the quality of engagement following the loss of status.

Moreover, we examine the temporal shift in the effort and quality of user engagement following the loss of status by extending the analysis to the monthly level. Consistent with the previous steps, we modified the dependent variable in Eq. (2) to $Log\#RevLen^{Month}$ and $Log\#RevVote^{Month}$ to capture monthly engagement effort and quality, respectively. The results are presented in Appendix B, and the monthly trends are illustrated in Fig. 5. Both effort and quality show a significant decline in the first three months after losing status, followed by a diminution of this decline in subsequent months, similar to the pattern observed for engagement frequency. However, the declines in effort and quality are less prolonged than the decline in frequency. This is likely because frequency captures overall disengagement, including those who stop participating, whereas effort and quality only capture the remaining active users who are more motivated to maintain their performance. These findings provide a more detailed illustration of the short- and long-term effects of status loss on user engagement.

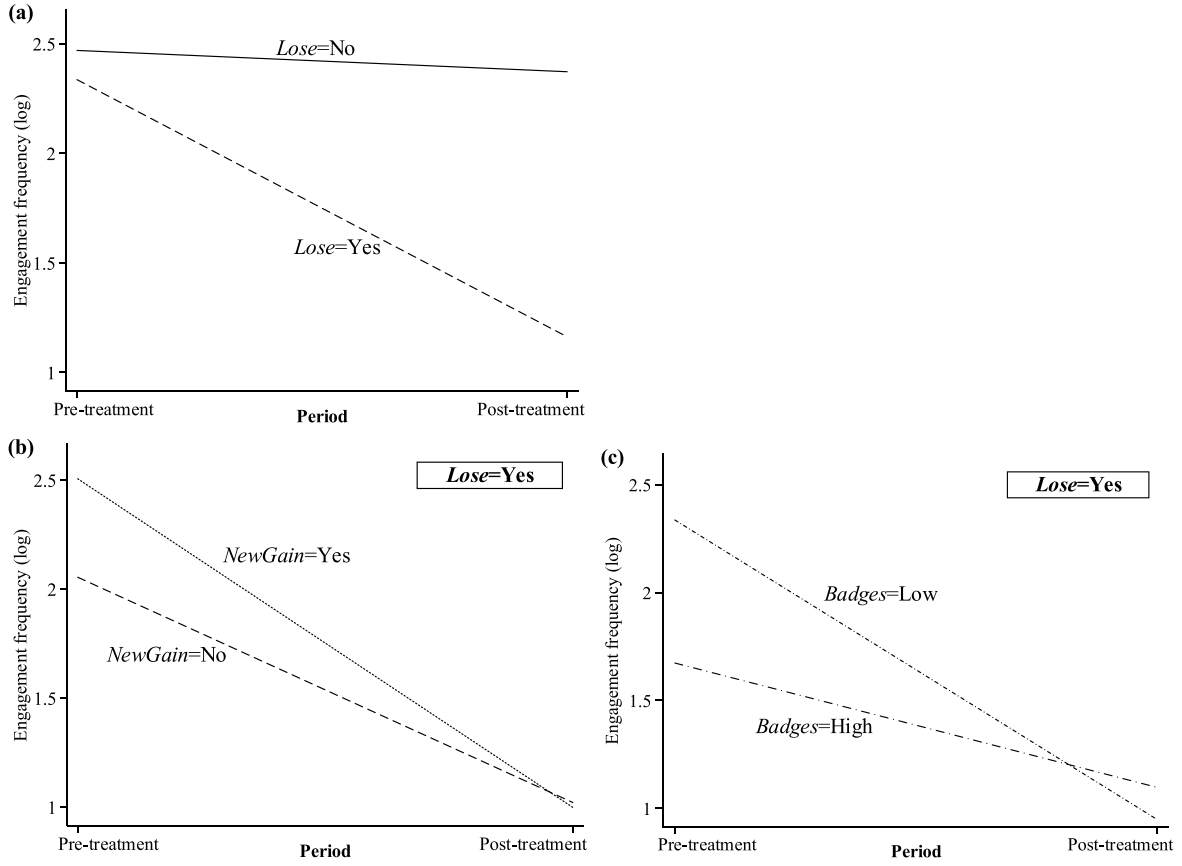


Fig. 3. Predictive margins of (a) the effect of status loss, (b) moderated by *NewGain*, and (c) moderated by *Badges*.

Table 6
Monthly effects of status loss on user engagement frequency.

	DV: $Log\#Rev_N^{Month}$	
	(1)	(2)
*Month*_ ×	*Post* × *Lose*	*Post* × *Lose* × *NewGain*
1	−0.311***	−0.324
	(0.079)	(0.176)
2	−0.450***	−0.440**
	(0.076)	(0.169)
3	−0.340***	−0.410**
	(0.072)	(0.158)
4	−0.285***	−0.050
	(0.069)	(0.152)
5	−0.196**	−0.115
	(0.070)	(0.150)
6	−0.090	−0.117
	(0.066)	(0.141)
7	−0.132*	0.002
	(0.067)	(0.145)
8	−0.039	−0.071
	(0.063)	(0.134)
9	0.046	0.104
	(0.061)	(0.130)
10	0.176**	0.093
	(0.060)	(0.124)
11	0.034	−0.032
	(0.060)	(0.121)
12	(base)	(base)
Control	YES	YES
City FE	YES	YES
N	22,704	22,704
R^2	0.134	0.138

Notes: Subvariables and sub-interaction terms derived from the multiple interaction terms are included in the control item.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

5.3. Directionality of changes in different behavior dimensions

To investigate the directional shifts in users' engagement behavior across various dimensions before and after experiencing status loss, we first group users into high, medium, and low engagement levels based on their ex ante (i.e., in 2016) performance in engagement frequency, effort, and quality, respectively. Subsequently, we perform regression analyses as follows: (1) regression with users grouped by ex ante engagement frequency, using engagement frequency as the dependent variable; (2) regression with users grouped by ex ante engagement effort, using engagement effort as the dependent variable; and (3) regression with users grouped by ex ante engagement quality, using engagement quality as the dependent variable. We then examine the regression coefficients of the nine treatment effects *Post* × *Lose*, as depicted in Table 8. The findings reveal that in each focal dimension, as users' ex ante level of engagement in that dimension increases, the negative effect of status loss gradually intensifies, thereby supporting H6a, H6b, and H6c.

To further validate our proposed directional changes across engagement dimensions, we cross-examine the relationships between regression test dimensions and grouping dimensions. For instance, when users are categorized based on ex ante engagement frequency, we regress on engagement effort and quality as dependent variables, and likewise for other groupings. Table 9 presents the regression coefficients for *Post* × *Lose*. The results indicate that when users are categorized into low, medium, and high levels based on ex ante engagement frequency and quality, no discernible directional change is observed in users' behavior in the non-focal dimension. Similarly, when users are categorized based on ex ante effort, the directional changes in engagement frequency and quality are not as evident as observed when users are categorized into levels based on the corresponding dimensions.

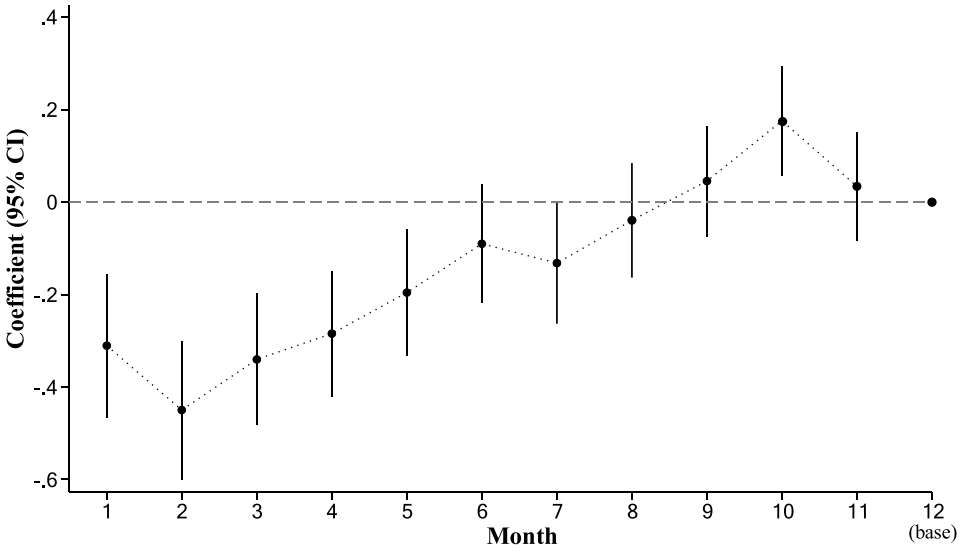


Fig. 4. Monthly trends in engagement frequency.

Table 7
Effects of status loss on user engagement effort and quality.

DV:	*Log#RevLen*	*Log#RevVote*
	(1)	(2)
Post	−0.288***	−0.217***
	(0.069)	(0.033)
Lose	−0.059	−0.079*
	(0.054)	(0.040)
Post × *Lose*	−2.401***	−0.505***
	(0.164)	(0.050)
NewGain	0.096	−0.013
	(0.123)	(0.048)
Badges	−0.086	−0.001
	(0.046)	(0.023)
CumRevs	0.087	0.060*
	(0.067)	(0.030)
AvgRating	−0.135	−0.022
	(0.074)	(0.035)
SocialReach	0.057	0.207***
	(0.038)	(0.019)
Tenure	0.089**	−0.008
	(0.028)	(0.012)
City FE	YES	YES
N	1892	1892
R^2	0.318	0.398

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

These results further validate our hypothesis of directional changes in users' engagement performance across each dimension following status loss.

5.4. Endogeneity tests

The observed effects of status loss on user engagement frequency, effort, and quality may be a result of behavioral trends that originated in earlier periods. This raises potential endogeneity concerns, particularly regarding the reciprocal relationship between engagement and status loss. To address these issues, we employ alternative matching methods and extend the analysis to include engagement metrics over a longer pre-treatment period, thereby capturing underlying trends more comprehensively.

First, we apply coarsened exact matching (CEM), a method that improves covariate balance by coarsening variables into exact categories [13], to re-match the treatment and control groups. Covariates include user engagement frequency, contribution effort, content quality from the three pre-treatment years (2014, 2015, and 2016), number of badges, and user tenure, ensuring better comparability of engagement trends and platform experience across groups. Cutpoints for each variable are determined using Sturges' rule [31]. The re-estimated results, presented in Table 10, indicate significant negative effects of status loss on engagement frequency, effort, and quality. Additionally, we conduct alternative CEM processes by specifying finer covariate cutoffs, yielding results consistent with our main findings. Further details and results are provided in Appendices C and D.

Second, we apply the generalized synthetic control method (GSCM), which generates counterfactual outcomes for treated units by combining untreated units, accounting for both observed and unobserved variations over time and across units [42]. The method addresses time-varying confounders by allowing individual fixed effects to change over time. We construct a user-year level panel beginning in 2014, with the status loss event occurring in 2017. GSCM captures the trends of each focal variable during the pre-treatment period (2014–2016). Since GSCM requires a longer pre-treatment period to construct synthetic controls, the number of observations in the treatment group is notably reduced. We incorporate two-way fixed effects (user and year) into the model and remove pre-treatment periods shorter than 3 years. The re-estimated results, presented in Table 11, show significant negative effects of status loss, consistent with our main findings.

Third, we apply a two-stage approach similar to the method used in Gao et al. [10] to account for potential behavioral trends and mitigate endogeneity concerns. In the first stage, Gao et al. [10] estimated group and period fixed effects using the subsample of untreated observations. In the second stage, they subtracted the estimated group and period effects from the observed outcomes and then regress the adjusted outcomes on treatment status to estimate the causal effect. Similarly, we adopt a two-stage approach to estimate the effect of status loss on user engagement while controlling for potential behavioral trends and endogeneity. In the first stage, we estimate the pre-treatment trend (2014–2016) as a function of tenure, individual user dummies, and their interactions. This captures individual-specific effects and other relevant variables influencing user engagement behaviors before the event, thus controlling for underlying engagement patterns and behavioral tendencies, similarly to how Gao et al. [10] account for baseline individual unobserved features. Using the estimated coefficients, the observed values of the independent variables, and the estimated individual fixed effects, we predict post-treatment engagement for each user. In the second stage, we incorporate these predicted engagement variables—frequency, effort, and quality—into our main model, along with tenure

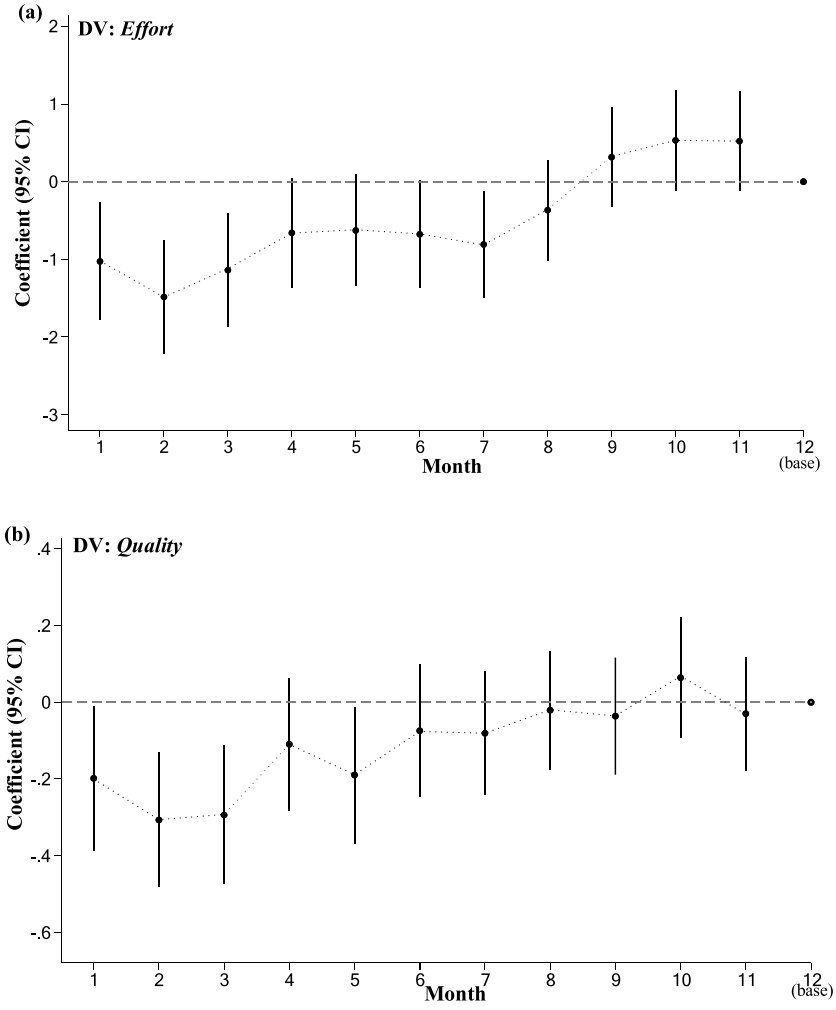


Fig. 5. Monthly trends in engagement (a) effort and (b) quality.

Table 8
Effects of status loss on different levels of users across various engagement dimensions.

	Focal engagement dimension level		
	Low	Medium	High
Log#Rev_N	−1.021***	−1.104***	−1.625***
	(0.109)	(0.106)	(0.115)
Log#RevLen	−1.668***	−2.454***	−3.146***
	(0.267)	(0.278)	(0.311)
Log#RevVote	−0.341***	−0.498***	−0.797***
	(0.055)	(0.077)	(0.103)

Notes: The results displayed are the coefficients of *Post* × *Lose*.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

dummies to further control for platform experience trends. By including this trend variable, we effectively deduct the pre-treatment behavioral trend from the post-treatment engagement outcomes, isolating the impact of status loss. This is analogous to Gao et al. [10], where adjusted outcomes are regressed on treatment status to isolate the causal effect.

The estimation is performed on the post-treatment period. Table 12 presents the results, with *Predict_Trend* representing the predicted values for the focal engagement dimensions. For comparison, we also show the results without *Predict_Trend*. After incorporating *Predict_Trend*, the

Table 9
Results of mixing between grouping and dependent variable dimensions.

DV	Engagement frequency		
	Low	Medium	High
Log#RevLen	−2.962***	−1.694***	−2.115***
	(0.284)	(0.304)	(0.289)
Log#RevVote	−0.512***	−0.385***	−0.505***
	(0.103)	(0.087)	(0.078)
	Engagement effort		
	Low	Medium	High
Log#Rev_N	−0.813***	−1.191***	−1.247***
	(0.113)	(0.114)	(0.111)
Log#RevVote	−0.405***	−0.483***	−0.650***
	(0.082)	(0.082)	(0.095)
	Engagement quality		
	Low	Medium	High
Log#Rev_N	−1.113***	−1.069***	−1.060***
	(0.105)	(0.120)	(0.117)
Log#RevLen	−2.456***	−1.853***	−2.922***
	(0.258)	(0.296)	(0.314)

Notes: The results displayed are the coefficients of *Post* × *Lose*. The leftmost column shows the dependent variables.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Table 10
Results based on coarsened exact matching (CEM).

DV:	*Log#Rev_N*	*Log#RevLen*	*Log#RevVote*
Post	−0.484***	−0.390**	−0.231***
	(0.082)	(0.146)	(0.060)
Lose	−0.590***	−0.044	−0.093
	(0.074)	(0.093)	(0.058)
Post × *Lose*	−0.825***	−2.207***	−0.464***
	(0.122)	(0.298)	(0.082)
Control variables	YES	YES	YES
City FE	YES	YES	YES
N	686	686	686
R^2	0.477	0.329	0.371

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Table 11
Results based on generalized synthetic control method (GSCM).

DV:	*Log#Rev_N*	*Log#RevLen*	*Log#RevVote*
Post × *Lose*	−0.795***	−0.139***	−0.336***
	(0.052)	(0.028)	(0.030)
User FE	YES	YES	YES
Year FE	YES	YES	YES
Number of treated users	234	234	234
Number of control users	1852	1852	1852

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Table 12
Results with and without detrending.

DV:	*Log#Rev_N*		*Log#RevLen*		*Log#RevVote*	
Detrend	NO	YES	NO	YES	NO	YES
Lose	−1.405***	−0.825***	−2.552***	−0.139***	−0.598***	−0.173***
	(0.066)	(0.064)	(0.176)	(0.041)	(0.050)	(0.049)
Predict_Trend		0.167***		0.450***		0.333***
		(0.027)		(0.028)		(0.037)
Control variables	YES		YES		YES	
Tenure dummies	YES		YES		YES	
City FE	YES		YES		YES	
N	930	727	930	727	930	727
R^2	0.425	0.371	0.281	0.401	0.365	0.532

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

effects of status loss on engagement frequency, effort, and quality are reduced, suggesting the presence of behavioral trends. Nonetheless, the results remain significant, indicating that our findings are robust against potential endogeneity concerns.

Fourth, we check for endogeneity using the Durbin–Wu–Hausman test, following an approach similar to that of Bhattacharyya et al. [5]. This test addresses potential unaccounted factors that might introduce endogeneity related to status loss in our context [2,5]. Specifically, it examines whether users' engagement is affected by status loss even when potential unaccounted-for factors are present. In the first stage, we use users' pre-treatment engagement metrics—frequency, effort, and quality—as explanatory variables in a regression model to predict their likelihood of losing status (i.e., *Lose*). This approach isolates the variation in status loss attributable to prior engagement metrics. The residuals from this regression (*Lose_res*) capture the unaccounted factors influencing status loss that are not explained by the pre-treatment metrics. In the second stage, we include *Lose_res* as an additional explanatory variable in our main model to test whether the unaccounted factors affect users' post-treatment engagement. Significant coefficients for *Lose_res* would indicate the presence of unaccounted endogeneity in our analysis, whereas insignificant coefficients suggest its absence [2,5]. The results of each stage are reported in Appendix E. The coefficients for *Lose_res* are nonsignificant for all engagement metrics—frequency, effort, and quality—indicating no evidence of unaccounted endogeneity. Additionally, the coefficients for *Lose* remain negative and significant across all models, confirming the robustness of our findings and supporting the conclusion that losing status negatively impacts user engagement.

5.5. Robustness checks

In our sample, while all users had posted reviews in 2016 (ex ante), some users may have discontinued their platform use thereafter, resulting in a loss of status and a subsequent decrease in review frequency between the two years. Additionally, the engagement effort and quality of users who ceased posting reviews in 2017 (ex post) may exhibit bias, as non-posting users were treated as contributing zero effort and quality, potentially inflating the impact of the loss-of-status factor. Therefore, we excluded users from our sample who did not post a review in 2017. The re-estimated results are presented in Appendices F and G, which align with the original results.

For our primary analysis, we selected users from the Yelp open dataset whose reviews accounted for more than 70 % of their total reviews. We adjusted this criterion to a more stringent 80 %[2] and re-estimated our results, as shown in Appendices H and I. These results are consistent with the original findings, reinforcing the robustness of our conclusions.

Our original model controls for city effects. To further mitigate unobserved heterogeneity among users, we incorporate individual fixed effects into our model. The results, detailed in Appendices J and K, are consistent with the original findings, thereby confirming the robustness of our results.

In our main results, we measure user engagement quality using the number of votes received. Additionally, to further validate the robustness of our findings, we adopt the method proposed by Korfiatis et al. [15]. Specifically, we use the proportion of useful votes to total votes received by a review as the metric for engagement quality. The re-estimated results, presented in Appendix L, confirm the consistency with our original findings.

Yelp users can add friends and build social networks on the platform, with more experienced users likely having larger or more developed networks. Since Elite badges are awarded by community managers from a pool of nominees, experienced reviewers might know how to game the system or use external connections to influence managers. To address this potential bias, we divide users into two groups based on tenure and estimate the results for each of the two groups. The results, detailed in Appendix M, align with our original findings. Additionally, we observed a stronger effect of status loss events among less experienced users in the

[2] We avoided a stricter screening criterion (e.g., 90% or 100%) to prevent a small sample size, given the rare event issue. Opting for a more moderate criterion ensured ample sample size for reliable regression analysis.

analysis.

6. General discussion

The awarding of virtual status to users is a crucial mechanism for user identification within the current ecology of online platforms. Previous research has linked changes in user engagement to the granting and enhancement of virtual status, emphasizing its role as a positive reinforcer of user behavior. However, few studies have investigated the negative impact of virtual status systems on user engagement. This study is the first to examine the role of the virtual status system as a negative reinforcer meticulously, focusing specifically on the effects of virtual status loss on users' engagement behavior.

6.1. Theoretical contributions

This study makes a significant contribution to reinforcement theory by examining the effects of status loss as a form of negative reinforcement on online user engagement in terms of frequency, effort, and quality. While extant literature has connected reinforcement theory to online user behavior, it has predominantly emphasized the reinforcing effects of positive factors. In contrast, this research extends reinforcement theory by exploring the inhibitory effects as a form of negative reinforcement. This expansion addresses the literature's unbalanced focus on positive reinforcement, revealing a dual-impact mechanism of status change and enhancing the theory's explanatory power in online interaction contexts.

This study leverages accountability theory to expand the theoretical understanding of how status loss impacts engagement behavior. Unlike private, consumption-based status systems where status loss is typically explained through emotional mechanisms such as frustration and social discomfort [3], this study emphasizes that, in public, reputation-based virtual status loss, perceived accountability plays a central role. Our research offers new theoretical insights into how virtual status loss reduces user engagement, underscoring distinct behavioral mechanisms of status loss in non-monetary (versus monetary) and public (versus private) virtual status systems.

This study offers a fresh perspective on how status loss affects user engagement motivation by examining two moderating variables, status acquisition experience and user engagement preferences, contributing to a deeper understanding of the varied engagement behavior influenced by individual differences. The findings show that users with more status experiences are less affected by status loss. Additionally, by examining the directional drift of user behavior, this study reveals that personal preferences significantly influence how users adjust their behavior following status loss. This discovery expands the existing literature on status effects by providing new insights into the multi-layered structure of user engagement motivation. It suggests that users may develop attachments to specific behavioral dimensions on the platform, and these preferences can lead to an uneven decline in motivation following status loss.

The study contributes further by analyzing the temporal dynamics of negative reinforcement on user engagement. By examining user behavior trends at various successive time points, the research finds that users exhibit a rapid negative feedback response to status loss, most pronounced within the first three months, which stabilizes after six months. This temporal analysis enriches the literature on the dynamic effects of status changes on online user behavior, elucidating how user engagement evolves in response to status-related events.

6.2. Practical implications

A key finding of this study is the detrimental effect of virtual status systems on platform user engagement. Previous research has predominantly highlighted the positive outcomes associated with status gains, such as badges, which have been shown to enhance online word of mouth and knowledge exchange. This has led platforms to implement more intricate status-awarding mechanisms, with the expectation of fostering more active engagement and higher-quality contributions from users. However, this focus has largely overlooked the potential adverse consequences of such mechanisms for user engagement.

Our findings indicate that the negative impact of status loss affects user performance significantly across multiple dimensions of engagement. This observation validates the existence of risks associated with status-awarding systems. Consequently, it is imperative for platforms to acknowledge not only the positive effects of these systems but also their potential to influence user engagement negatively when users experience status loss or fail to attain status.

Particularly concerning is the Matthew effect-like phenomenon observed, where users who are initially more active receive positive reinforcement, while less active users experience negative reinforcement. This disparity can result in less active users lacking both the channels and the motivation to engage effectively on the platform, potentially leading to their departure. Such an outcome undermines the platform's goal of fostering a thriving community.

We must acknowledge that if a platform evaluates user contributions based on objective criteria, status may be revoked when users fail to meet those standards. Even if platform managers understand that status loss can harm user motivation, it may be unavoidable in some cases. In organizational management, early identification of employee behavior and timely feedback are emphasized as essential for enhancing performance [22,24]. Similarly, platforms should implement supportive mechanisms to address status deprivation or non-award situations effectively. Before status loss occurs, platforms could notify users that their recent engagement might lead to status removal, helping them recognize the potential to reverse any decline in motivation. After status loss, targeted encouragement and recommendations can help mitigate negative effects and maintain user engagement. These could include providing users with feedback on the reasons for their status loss, suggesting areas for improved engagement, offering consolation and recognition for past achievements, or awarding lower-level status badges. As Bhattacharyya et al. [5] noted, recognition serves as a cost-effective approach to fostering user contributions. These measures can help prevent the negative repercussions of status systems and support the platform's goal of enhancing community engagement.

Furthermore, the study's finding that newly gained status users exhibit more pronounced negative responses to status loss suggests that status system designers should adopt a more lenient approach for these users. Specifically, lowering the evaluation criteria for newly gained status users could help mitigate the adverse effects of status loss. Additionally, our findings highlight the necessity of timely interventions to counteract the immediate negative impacts of status loss. Implementing short-term encouragement and engagement recommendations can effectively mitigate these effects.

Last, our findings indicate that the contribution dimensions affected by the status loss (e.g., frequency, effort, and quality) align with their engagement preferences. This provides more precise guidance for platform managers to tailor intervention strategies based on these preferences. Platforms can strategically adjust the criteria for status awards based on the dimensions they prioritize. By lowering the thresholds for key engagement dimensions, platforms can better promote community prosperity in line with specific objectives.

6.3. Limitations and future research

While this study provides valuable insights into the effects of status loss as a form of negative reinforcement on online user engagement, several limitations should be acknowledged.

First, this study focuses on a specific online platform, which may limit the generalizability of the findings. The status systems and rules for awarding status on other platforms may differ significantly from those on Yelp. For instance, Tabelog (https://tabelog.com/), one of Japan's

largest restaurant review platforms, awards users gold, silver, and bronze badges based on their engagement over the previous year. Given the hierarchical differences in status on such platforms, the psychomotivational changes users experience upon losing status are likely more multifaceted than those of Yelp users. This includes tier differences in the lost status and differences in the span of status tiers (e.g., from Gold to Silver or Gold to Bronze). Drawing on the findings and theoretical frameworks (reinforcement and accountability theories) of this paper, future research could investigate the impact of status loss on user engagement under more complex status-awarding systems. Additionally, adopting the analytical approaches used in this paper, future research could explore the dynamic impact of status loss over time.

Second, this study's examination of negative reinforcement focuses on the context of status loss. However, other forms of negative reinforcement, such as increased difficulty in gaining status, reforms to the status system (e.g., introducing more tiers or categories of badges), or even the elimination of the status system, may also affect user engagement. Future research could investigate these additional forms of negative reinforcement, particularly in the context of relevant changes to platform status systems, to provide a more comprehensive understanding of negative reinforcement in online environments.

Finally, this study primarily examines individual differences based on users' historical engagement with the platform. However, users' reactions to status loss events are also correlated with personality, character, preferences, and cultural background. Future research can explore the impact of negative reinforcement events on user engagement by collecting more diverse individual user characteristics through laboratory or online experimental approaches. This would allow a more thorough consideration of user heterogeneity, enhancing our understanding of how personal factors influence reactions to status loss and other forms of negative reinforcement.

7. Conclusions

The strategic management of virtual status systems within online communities plays a crucial role in sustaining user engagement, especially considering the profound influence these systems have on user behavior. This study sheds light on the often-overlooked negative consequences of virtual status loss for user engagement within online platforms. By analyzing behavioral changes following status loss, we observed a significant decline in user participation, particularly immediately after the loss. The impact varies depending on the users' prior experiences with status, with those losing newly gained status experiencing more substantial declines in engagement. This highlights the critical need for platforms to recognize the dual nature of virtual status systems, where both the positive reinforcement of status gains and the negative reinforcement of status loss play pivotal roles in shaping user behavior.

Our findings highlight the importance of designing status systems that mitigate the adverse effects of status loss. Platforms should develop strategies to maintain user motivation and engagement, such as providing support and alternative incentives for users who lose their status. By doing so, they can foster a healthier and more active online community. This research contributes to a more nuanced understanding of virtual status systems and offers valuable insights for optimizing user engagement strategies. Future research should continue to explore these effects across diverse platform contexts and user experiences, further enriching our comprehension of the complex dynamics at play in virtual status systems.

CRediT authorship contribution statement

Ruonan Zhang: Writing – review & editing, Project administration, Formal analysis, Conceptualization. **Zhaoyang Yu:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A
Quarterly and semiannual effects of status loss on user engagement frequency.

	DV : $Log\#Rev_N^{quater}$		DV : $Log\#Rev_N^{semiannual}$
Quater ×	*Post* × *Lose*	***Semiannual*** ×	*Post* × *Lose*
Q1	−0.437***	**FirstHalf**	−0.292***
	(0.052)		(0.038)
Q2	−0.260***	**SecondHalf**	(base)
	(0.046)		
Q3	−0.112***		
	(0.039)		
Q4	(base)		
Control	YES	Control	YES
City FE	YES	City FE	YES
N	22,704	*N*	22,704
R^2	0.114	R^2	0.108

Notes: Subvariables and sub-interaction terms derived from the multiple interaction terms are included in the control item. *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix B
Monthly effects of status loss on user engagement effort and quality.

DV:	$Log\#RevLen^{Month}$	$Log\#RevVote^{Month}$
Month		
1	−1.030**	−0.203*
	(0.384)	(0.096)
2	−1.534**	−0.327***
	(0.372)	(0.089)
3	−1.143**	−0.299**
	(0.371)	(0.092)
4	−0.662	−0.114
	(0.360)	(0.088)
5	−0.631	−0.195*
	(0.365)	(0.091)
6	−0.678	−0.074
	(0.353)	(0.088)
7	−0.809*	−0.081
	(0.350)	(0.082)
8	−0.367	−0.021
	(0.330)	(0.079)
9	0.315	−0.037
	(0.327)	(0.078)
10	0.543	0.063
	(0.330)	(0.080)
11	0.522	−0.031
	(0.325)	(0.075)
12	(base)	(base)
Control	YES	YES
City FE	YES	YES
N	14,905	14,905
R^2	0.127	0.133

Notes: Subvariables and sub-interaction terms derived from the multiple interaction terms are included in the control item.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix C
Results based on coarsened exact matching with equally spaced cutpoints.

DV:	*Log#Rev_N*	*Log#RevLen*	*Log#RevVote*
Post	−0.469***	−0.226**	−0.139***
	(0.049)	(0.073)	(0.033)
Lose	−0.709***	−0.005	−0.011
	(0.056)	(0.058)	(0.042)
Post × *Lose*	−0.723***	−2.175***	−0.496***
	(0.081)	(0.210)	(0.058)
Control variables	YES	YES	YES
City FE	YES	YES	YES
N	1522	1522	1522
R^2	0.499	0.294	0.341

Notes: The covariates used in CEM include engagement frequency, effort, and quality from 2014, 2015, and 2016, as well as the number of badges and user tenure. All covariates are coarsened into 10 equally spaced cutpoints.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix D
Results based on coarsened exact matching with percentile-based cutpoints.

DV:	*Log#Rev_N*	*Log#RevLen*	*Log#RevVote*
Post	−0.391**	−0.045	−0.116
	(0.121)	(0.123)	(0.083)
Lose	−0.343	0.335	0.196
	(0.195)	(0.202)	(0.141)
Post × *Lose*	−1.721***	−3.561***	−1.001***
	(0.259)	(0.656)	(0.191)
Control variables	YES	YES	YES
City FE	YES	YES	YES
N	262	262	262
R^2	0.632	0.557	0.740

Notes: The covariates used in CEM include engagement frequency, effort, and quality from 2014, 2015, and 2016, as well as the number of badges and user tenure. The cutpoints for each covariate are based on the 25th, 50th, and 75th percentiles from the treatment group sample.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix E
Results of the Durbin–Wu–Hausman test.

Stage 1: Dependent variable–*Lose*.			
Independent variables:	*Log#Rev_N*	*Log#RevLen*	*Log#RevVote*
Coefficient	−0.137*** (0.021)	0.047 (0.033)	−0.084** (0.026)

Stage 2: Results with engagement frequency, effort, and quality as dependent variables.					
Dependent variables:	***Log#Rev_N***		***Log#RevLen***		***Log#RevVote***
Lose	−1.457***		−1.787***		−0.842***
	(0.198)		(0.500)		(0.194)
Lose_res	0.102		−0.672		0.292
	(0.214)		(0.531)		(0.209)
Controls	YES		YES		YES

Notes: Stage 1 uses the sample from the pre-treatment period, while Stage 2 uses the sample from the post-treatment period.
*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix F
Status loss impact on engagement frequency (for users with reviews both pre- and post-treatment).

	DV: *Log#Rev_N*		
	(1)	(2)	(3)
Post	0.007	0.072	−0.038
	(0.040)	(0.051)	(0.064)
Lose	−0.277***	−0.373***	−0.277***
	(0.047)	(0.055)	(0.062)
Post × *Lose*	−0.565***	−0.423***	−0.845***
	(0.064)	(0.077)	(0.107)
Post × *Lose* × *NewGain*		−0.486***	
		(0.134)	
Post × *Lose* × *Badges*			0.115**
			(0.038)
Post × *NewGain*		−0.167*	
		(0.075)	
Lose × *NewGain*		0.328***	
		(0.093)	
Post × *Badges*			0.033
			(0.025)
Lose × *Badges*			0.002
			(0.027)
NewGain	0.276***	0.308***	0.285***
	(0.056)	(0.069)	(0.056)
Badges	−0.064**	−0.074**	−0.105***
	(0.023)	(0.023)	(0.027)
CumRevs	0.055	0.092*	0.078
	(0.040)	(0.041)	(0.041)
AvgRating	−0.178***	−0.164***	−0.180***
	(0.041)	(0.039)	(0.041)
SocialReach	0.065***	0.059**	0.063***
	(0.019)	(0.019)	(0.019)
Tenure	−0.031*	−0.034**	−0.032*
	(0.013)	(0.013)	(0.013)
City FE	YES	YES	YES
N	1686	1686	1686
R^2	0.270	0.288	0.281

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix G
Status loss impact on engagement effort/quality (for users with reviews both pre- and post-treatment).

DV:	*Log#RevLen*	*Log#RevVote*
Post	−0.005	−0.164***
	(0.023)	(0.032)
Lose	−0.000	−0.069
	(0.035)	(0.040)
Post × *Lose*	−0.114**	−0.123**
	(0.038)	(0.047)
NewGain	−0.024	−0.022
	(0.046)	(0.048)
Badges	0.004	0.017
	(0.017)	(0.023)
CumRevs	0.040	0.056
	(0.026)	(0.030)

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Appendix G (*continued*)

DV:	*Log#RevLen*	*Log#RevVote*
AvgRating	−0.026	−0.003
	(0.031)	(0.035)
SocialReach	0.078***	0.234***
	(0.014)	(0.020)
Tenure	−0.008	−0.029*
	(0.011)	(0.012)
City FE	YES	YES
N	1686	1686
R^2	0.139	0.387

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix H

Status loss impact on engagement frequency (for users with more than 80 % of their total reviews published by Yelp's open dataset).

	DV: *Log#Rev_N*		
	(1)	(2)	(3)
Post	−0.159**	−0.090	−0.155
	(0.056)	(0.069)	(0.094)
Lose	−0.342***	−0.487***	−0.294***
	(0.058)	(0.069)	(0.075)
Post × *Lose*	−1.053***	−0.944***	−1.376***
	(0.079)	(0.096)	(0.140)
Post × *Lose* × *NewGain*		−0.370*	
		(0.166)	
Post × *Lose* × *Badges*			0.161**
			(0.052)
Post × *NewGain*		−0.169	
		(0.105)	
Lose × *NewGain*		0.424***	
		(0.112)	
Post × *Badges*			0.008
			(0.034)
Lose × *Badges*			−0.037
			(0.038)
NewGain	0.323***	0.309***	0.325***
	(0.076)	(0.089)	(0.077)
Badges	−0.064	−0.072*	−0.095*
	(0.034)	(0.034)	(0.039)
CumRevs	0.077	0.109	0.099
	(0.056)	(0.057)	(0.057)
AvgRating	−0.213***	−0.198***	−0.211***
	(0.052)	(0.049)	(0.051)
SocialReach	0.047	0.043	0.044
	(0.025)	(0.025)	(0.025)
Tenure	0.006	0.003	0.005
	(0.018)	(0.018)	(0.018)
City FE	YES	YES	YES
N	1244	1244	1244
R^2	0.440	0.451	0.447

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix I

Status loss impact on engagement effort/quality (for users with more than 80 % of their total reviews published by Yelp's open dataset).

DV:	*Log#RevLen*	*Log#RevVote*
Post	−0.348***	−0.206***
	(0.089)	(0.041)
Lose	−0.000	−0.050
	(0.068)	(0.048)
Post × *Lose*	−2.381***	−0.505***
	(0.200)	(0.060)
NewGain	0.289	0.002
	(0.158)	(0.054)
Badges	−0.059	−0.008
	(0.064)	(0.025)
CumRevs	0.166	0.069
	(0.088)	(0.036)
AvgRating	−0.145	−0.006
	(0.090)	(0.042)

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Appendix I (*continued*)

DV:	*Log#RevLen*	*Log#RevVote*
SocialReach	0.042	0.197***
	(0.048)	(0.024)
Tenure	0.090*	−0.001
	(0.036)	(0.014)
City FE	YES	YES
N	1244	1244
R^2	0.327	0.403

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix J
Status loss impact on engagement frequency (with user's individual fixed effect).

	DV: *Log#Rev_N*		
	(1)	(2)	(3)
Post × *Lose*	−1.131***	−1.040***	−1.405***
	(0.061)	(0.075)	(0.111)
Post × *Lose* × *NewGain*		−0.257*	
		(0.128)	
Post × *Lose* × *Badges*			0.113**
			(0.038)
Post × *NewGain*		0.167	
		(0.097)	
Post × *Badges*			−0.065*
			(0.028)
Badges	0.064	0.007	0.229*
	(0.050)	(0.057)	(0.092)
CumRevs	−0.664***	−0.694***	−0.694***
	(0.075)	(0.089)	(0.083)
AvgRating	0.223*	0.246**	0.248**
	(0.086)	(0.089)	(0.088)
User FE	YES	YES	YES
N	1892	1892	1892
R^2	0.800	0.801	0.802

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix K
Status loss impact on engagement effort/quality (with user's individual fixed effect).

DV:	*Log#RevLen*	*Log#RevVote*
Post × *Lose*	−2.392***	−0.497***
	(0.163)	(0.049)
Badges	−0.308*	−0.262***
	(0.133)	(0.040)
CumRevs	0.169	0.165**
	(0.201)	(0.061)
AvgRating	−0.270	−0.101
	(0.232)	(0.070)
Tenure	−0.579*	−0.345***
	(0.229)	(0.069)
User FE	YES	YES
N	1892	1892
R^2	0.653	0.788

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix L
Measurement of engagement quality in terms of the ratio of useful votes.

DV:	*UsefulVoteRatio*
Post	−0.033***
	(0.007)
Lose	0.001
	(0.008)
Post × *Lose*	−0.121***
	(0.013)
NewGain	−0.005
	(0.010)
Badges	−0.004

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Appendix L (*continued*)

DV:	*UsefulVoteRatio*
	(0.004)
CumRevs	0.009
	(0.006)
AvgRating	−0.009
	(0.007)
SocialReach	0.025***
	(0.003)
Tenure	0.003
	(0.002)
City FE	YES
N	1892
R^2	0.242

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

Appendix M

Results based on users grouped by tenure.

DV:	*Log#Rev_N*		*Log#RevLen*		*Log#RevVote*	
Tenure:	Low	High	Low	High	Low	High
Post	−0.083	−0.012	−0.331*	−0.160	−0.318***	−0.168***
	(0.082)	(0.060)	(0.142)	(0.084)	(0.066)	(0.048)
Lose	−0.285***	−0.271***	0.074	−0.189	−0.087	−0.076
	(0.061)	(0.076)	(0.064)	(0.097)	(0.048)	(0.070)
Post × *Lose*	−1.342***	−0.916***	−3.083***	−1.904***	−0.585***	−0.451***
	(0.100)	(0.088)	(0.260)	(0.216)	(0.076)	(0.072)
NewGain	0.308**	0.119	0.134	0.063	0.055	0.006
	(0.100)	(0.089)	(0.190)	(0.190)	(0.076)	(0.074)
Badges	−0.149*	−0.099**	−0.070	−0.099	0.077	−0.009
	(0.064)	(0.033)	(0.118)	(0.059)	(0.051)	(0.030)
CumRevs	0.086	0.114	0.104	0.162	0.075*	0.048
	(0.053)	(0.076)	(0.091)	(0.110)	(0.036)	(0.051)
AvgRating	−0.235***	−0.142	−0.094	−0.155	−0.049	0.022
	(0.051)	(0.091)	(0.089)	(0.138)	(0.038)	(0.065)
SocialReach	0.068*	0.026	0.040	0.052	0.166***	0.250***
	(0.028)	(0.031)	(0.052)	(0.060)	(0.022)	(0.030)
Tenure	−0.039	−0.005	0.010	0.031	−0.034	−0.008
	(0.034)	(0.028)	(0.057)	(0.056)	(0.024)	(0.024)
City FE	YES	YES	YES	YES	YES	YES
N	955	926	955	926	955	926
R^2	0.507	0.364	0.419	0.270	0.418	0.404

*** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

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